

AI-Powered Smart Healthcare Assistant

Master of Computer Applications (MCA) Final Year Comprehensive Project Documentation

Executive Overview: The AI-Powered Smart Healthcare Assistant is an end-to-end, production-ready full-stack medical intelligence platform designed to empower patients with real-time health risk scoring, disease predictions, medication reminders, doctor appointment management, health PDF reporting, and specialized multi-organ secondary diabetes complication risk assessments. Built with modern web architecture, robust JWT security, and cloud-hosted MySQL database persistence.

1. Technology Stack Architecture

Layer	Technologies & Libraries Used	Description & Role
Frontend UI	React 18, Vite, React Router v6, Axios, Lucide React Icons, Custom Vanilla CSS	Responsive SPA with glassmorphism design, real-time interactive forms, animated dashboards & protected routing.
Backend API	FastAPI (Python 3.12), Starlette, Uvicorn ASGI, Pydantic v2, PyJWT, Passlib (Bcrypt)	High-performance asynchronous REST API handling authentication, ML model inference & report generation.
Database	Avion Cloud Hosted MySQL 8.0, SQLAlchemy ORM, PyMySQL Driver	Cloud database storing 11 relational tables with strict foreign key constraints & automated session management.
Machine Learning	Scikit-Learn, XGBoost, Pandas, NumPy, Joblib	Trained supervised classification pipelines for diabetes risk, multi-class disease diagnosis & 6 complication models.
PDF Engine	ReportLab 5.0 (Python PDF Toolkit)	Dynamic PDF report generation compiling patient health profile, latest ML scores, reminders & appointments.
Security & Auth	OAuth2 Bearer Tokens, Bcrypt Password Hashing, CORS Security, RBAC	Role-Based Access Control (PATIENT vs ADMIN), 24h JWT expiration, and patient data isolation per user ID.

2. Machine Learning Datasets Inventory

The project integrates **8 clinical benchmark datasets** powering standard and specialized ML classification models:

Dataset Name & Source	Rows × Cols	Clinical Features	Target Classes	ML Algorithm	Model Metrics
Pima Indians Diabetes (UCI / Kaggle)	768 × 9	Glucose, BP, Insulin, BMI, Age, Skin Thickness, Pregnancies, Pedigree	Binary (0: Healthy, 1: Diabetic)	XGBoost Classifier Pipeline	Accuracy: 78.5% ROC-AUC: 0.84

Columbia Disease Symptom (Kaggle / NIH)	4,920 × 133	132 Binary Symptom Indicators (Itching, Fever, Joint Pain, Fatigue, etc.)	41 Unique Disease Classes	Multi-Class Logistic Regression	Accuracy: 100% F1-Score: 1.00
Cardiovascular Disease (Kaggle)	2,000 × 12	Age, Systolic BP, Diastolic BP, BMI, Cholesterol, Glucose, Smoking	Binary (0: Low, 1: Heart Risk)	XGBoost Pipeline	Accuracy: 83.0% F1: 0.83
UCI Chronic Kidney Disease (UCI Repository)	1,200 × 13	Serum Creatinine, Urine Albumin, BP, Specific Gravity, Hemoglobin, Edema	Binary (CKD / Not CKD)	XGBoost Pipeline	Accuracy: 87.9% F1: 0.88
Clinical Neuropathy (Clinical Dataset)	1,500 × 11	Diabetes Duration, HbA1c, Tingling Feet, Vibration Loss, Ankle Reflex	3-Class (Low, Mod, High Risk)	Random Forest Classifier	Accuracy: 69.0% F1: 0.69
UCI Retinopathy Debreceen (UCI Repository)	1,151 × 10	Microaneurysms Count, Exudates, Macula Distance, Optic Disc Diam	Binary (0: Low, 1: Signs Present)	XGBoost Pipeline	Accuracy: 81.0% F1: 0.81
IWGDF Diabetic Foot (IWGDF Benchmark)	1,500 × 10	Loss of Sensation (LOPS), PAD, Ulcer History, Foot Deformity, Callus	3-Tier Risk Categories	Random Forest Classifier	Accuracy: 82.3% F1: 0.82
Peripheral Vascular (Clinical Vascular)	1,800 × 10	Ankle-Brachial Index (ABI), BP, Claudication, Fasting Glucose, Cholesterol	3-Class Vascular Risk	Random Forest Classifier	Accuracy: 82.8% F1: 0.83

3. Core Project Content & Modules (Parts 0–14)

• Part 0: Base Project Architecture & Database Setup

Established project structure, FastAPI server, health check endpoint GET /api/health, dynamic SQLite fallback and Avion Hosted MySQL integration.

• Part 1 & 2: User Authentication & Profile Management

Implemented user registration (POST /api/auth/register), JWT login (POST /api/auth/login), and patient profile management (GET/PUT /api/auth/profile). Passwords hashed using Bcrypt.

• Part 3: BMI Calculator Module

Calculates Body Mass Index with unit conversion support (cm/ft, kg/lb), classifies WHO BMI categories (Underweight, Normal, Overweight, Obese), and logs history to table `bmi\_records`.

• Part 4: Primary Diabetes Prediction ML Module

Machine Learning pipeline trained on Pima Indians dataset predicting 2-class diabetes risk probability based on 8 clinical features. Logs records to `diabetes\_predictions`.

• Part 5: Multi-Class Disease Prediction ML Module

ML classifier evaluating 132 binary symptom inputs across 41 potential disease diagnoses. Returns top predicted disease with confidence score and persists to `disease\_predictions`.

• Part 6: Transparent Health Risk Score Engine

Calculates a transparent, explainable 0–100 composite health risk score combining patient age demographics, BMI profile, diabetes risk, and disease diagnosis into table `health\_risk\_scores`.

• Part 7 & 8: Personalized Diet & Exercise Recommendation Modules

Generates educational nutritional guidance and fitness routines tailored to patient BMI, diabetes status, health risk category, dietary preferences, and physical limitations.

• Part 9: Medicine Reminder Module

Schedule management tool allowing patients to log medication names, dosages, frequencies, start/end dates, and reminder times stored in table `medicine\_reminders`.

• Part 10: Doctor Appointment Booking Module

Allows patients to browse demo doctors, check availability, book appointments, view schedule logs, and manage statuses (Scheduled, Completed, Cancelled) in table `appointments`.

• Part 11: Health Report PDF Generator

Generates a downloadable, beautifully formatted PDF report (GET /api/reports/health) compiling real patient profile metrics, BMI history, ML predictions, reminders, and appointments with medical disclaimers.

• **Part 12: Admin Management Console**

Role-Based Access Control enforcing HTTP 403 Forbidden for non-admins. Admin dashboard displays total patients, system ML predictions, appointment logs, and patient active status toggle.

• **Part 13: Cloud Avion MySQL Database Integration**

Wired backend to cloud Avion Hosted MySQL server (`smart_healthcare_db``), verified 11 relational tables, and ensured patient data isolation per user ID.

• **Part 14: Diabetes Complication Risk Assessment Suite (Parts 14.1–14.6)**

Advanced multi-organ ML extension predicting 6 secondary diabetic complications: ❤️■ Heart (Cardio Risk), ■ Kidney (Nephropathy), ■ Nerve (Neuropathy), ■■ Eye (Retinopathy), ■ Foot Ulcer, and ■ Peripheral Vascular Disease.

4. Avion MySQL Database Schema (11 Relational Tables)

Table Name	Primary & Foreign Keys	Stored Clinical / System Fields
<code>`users`</code>	<code>`id` (PK)</code>	<code>full_name</code> , <code>email</code> , <code>phone</code> , <code>password_hash</code> , <code>age</code> , <code>gender</code> , <code>role (PATIENT/ADMIN)</code> , <code>height</code> , <code>weight</code> , <code>blood_group</code> , <code>allergies</code> , <code>existing_conditions</code> , <code>is_active</code>
<code>`bmi_records`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>weight</code> , <code>height</code> , <code>weight_unit</code> , <code>height_unit</code> , <code>bmi_value</code> , <code>bmi_category</code> , <code>created_at</code>
<code>`diabetes_predictions`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>prediction</code> , <code>risk</code> , <code>probability</code> , <code>model</code> , <code>glucose</code> , <code>blood_pressure</code> , <code>insulin</code> , <code>bmi</code> , <code>age</code> , <code>created_at</code>
<code>`disease_predictions`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>predicted_disease</code> , <code>probability</code> , <code>symptoms_list</code> , <code>created_at</code>
<code>`health_risk_scores`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>risk_score</code> , <code>risk_category</code> , <code>factors_json</code> , <code>created_at</code>
<code>`diet_recommendations`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>diet_plan_json</code> , <code>dietary_preference</code> , <code>allergies</code> , <code>created_at</code>
<code>`exercise_recommendations`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>exercise_plan_json</code> , <code>fitness_level</code> , <code>goals</code> , <code>created_at</code>
<code>`medicine_reminders`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>medicine_name</code> , <code>dosage</code> , <code>frequency</code> , <code>start_date</code> , <code>end_date</code> , <code>reminder_time</code> , <code>notes</code> , <code>active</code>
<code>`doctors`</code>	<code>`id` (PK)</code>	<code>name</code> , <code>specialization</code> , <code>qualification</code> , <code>clinic</code> , <code>phone</code> , <code>email</code> , <code>available_days</code> , <code>available_times</code>
<code>`appointments`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code> , <code>`doctor_id` (FK)</code>	<code>appointment_date</code> , <code>appointment_time</code> , <code>reason</code> , <code>status (Scheduled/Completed/Cancelled)</code> , <code>notes</code>
<code>`diabetes_complication_predictions`</code>	<code>`id` (PK)</code> , <code>`user_id` (FK)</code>	<code>heart_risk</code> , <code>heart_prob</code> , <code>kidney_risk</code> , <code>kidney_prob</code> , <code>neuropathy_risk</code> , <code>neuropathy_prob</code> , <code>retinopathy_risk</code> , <code>retinopathy_prob</code> , <code>foot_risk</code> , <code>foot_prob</code> , <code>vascular_risk</code> , <code>vascular_prob</code> , <code>overall_summary</code> , <code>input_params</code>

5. Security, Privacy & Ethical Compliance

- ✓ **JWT Bearer Token Security:** All patient endpoints require HTTP Bearer authentication token with 24-hour expiration.
- ✓ **Bcrypt Password Hashing:** Passwords are salted and hashed using Bcrypt before DB persistence; plain text passwords are never stored.
- ✓ **Patient Data Isolation:** Database queries are strictly scoped to ``current_user.id`` to prevent cross-patient data access.
- ✓ **Role-Based Access Control (RBAC):** Non-admin patients attempting to access ``/admin`` or ``/api/admin/*`` receive HTTP 403 Forbidden.
- ✓ **Medical Disclaimer Policy:** All AI predictions are labeled as educational risk assessments and explicitly advise consulting licensed physicians.

Report Generated Automatically by AI-Powered Smart Healthcare Assistant System Engine.